

Homework 1

Due in five days · show all work · fictional coursework

Answer all four questions. Row operations must be labeled and final answers written in the notation requested. Work is graded against the published rubric.

USE THIS SYSTEM AND MATRIX A FOR QUESTIONS 1 AND 2:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

- 1 Solve the system** 8 pts
Solve the linear system by row reduction. Label every row operation and write the solution as a column vector.
- 2 Rank and nullity** 6 pts
Find the rank and nullity of the matrix A and give a basis for its null space.
- 3 Subspace proof** 8 pts
Prove that $W = \{ (x, y, z) : x + 2y - z = 0 \}$ is a subspace of $\mathbb{R}^3$.
- 4 Inverse** 8 pts
Find A^{-1} using the augmented matrix $[A \mid I]$ and verify with one product.